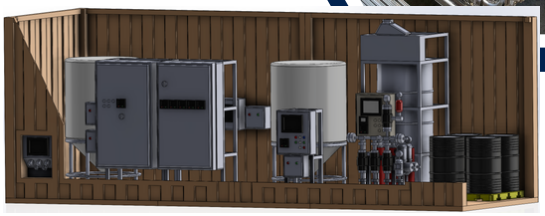
AxNano is the commercialization partner and distributor for General Atomic systems.

AxSure™

AxNano's AxSure™ system integrates an automated feed management system (AFS) with supercritical water oxidation (SCWO) to effectively eliminate contaminants resistant to conventional treatments. The AFS ensures efficient particle reduction and solid-liquid control, facilitating the handling of complex waste streams. By optimizing the SCWO process through continuous flow and homogeneous mixing, the system accommodates various waste types and reduces long-term environmental liabilities.

SUPERCRITICAL WATER OXIDATION TECHNOLOGY

When water is heated beyond 374°C and subjected to a pressure of 221 bar, it transitions into a supercritical state. This unique phase of water effectively decomposes organic contaminants into carbon dioxide, water, and inorganic salts.



AXNano

EFFECTIVE DESTRUCTION OF:

- Granular Activated Carbon (GAC)
- Ion Exchange Resin
- Semi-solids (Biosolids)
- High salt waste (RO Reject)
- Highly Concentrated PFAS
- Organic Compounds

DESTRUCTION OF PFAS CASE STUDY:

AFFF

- ⌀ 250-gallons
- ⌀ 30 mg/L
- ⌀ 99.99% DRE

*McDonough et al (2022)

BIOSOLIDS

- ⌀ 500-gallons
- ⌀ 4.9ug/L
- ⌀ 98.75% DRE

GAC

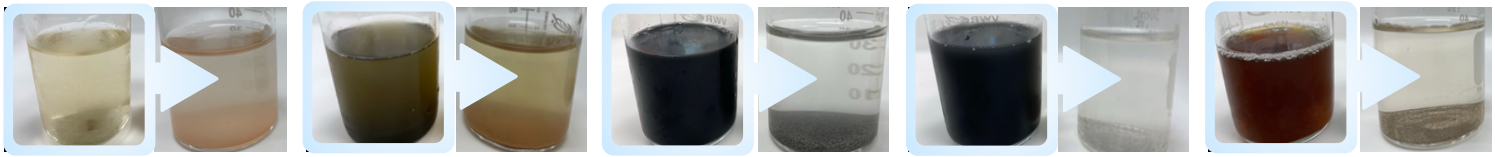
- ⌀ 150-gallons
- ⌀ 1.15 mg/L
- ⌀ 99.995% DRE

RESIN

- ⌀ 90-gallons
- ⌀ 134 ug/L
- ⌀ 98.52% DRE

RO REJECT

- ⌀ 650-gallons
- ⌀ 13.4 ug/L
- ⌀ 99.66% DRE



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